Graphite insert process:

1. Define:
    - Chamber Pressure p_c
    - Mass flow *mdot*
    - Hot-gas temperature T_g
    - Desired Burn-time t_{burn}
    - Throat diameter d_{throat}
2. Estimate Surface Heat Flux at the throat *q"*
    - Use Bartz Equation
    - Can be found using $r(dot)_{erosion}$

H_{abl} (enthalpy constant for graphite) = $5*10^7$ J/kg

$q" = \rho_{graphite}*H_{abl}*r(dot)_{erosion}$

3. Choose Graphite thermal properties
    - Conductivity *k*
        - Choose lower values for safety

k ~ 1500-2000 W/(m*K)

Use lower end ***k* ~ 1500 W/(m*K)**

- Thermal Diffusivity ***α***

$\rho_{graphite}$=2260 kg/m^3

$cp_{graphite}$ = 718.9-669.4 J/(kg*K)

$\alpha = k/(\rho_{graphite}*cp_{graphite})$

For low specific heat estimate:

= 1500/(2260)*(669.4) W*m^3*kg*K/m*K*kg*J

$\boldsymbol{\alpha_{low} = 9.92*10^{-4}\ m^2/s}$

For high specific heat estimate:

= 1500/(2260)*(718.9) W*m^3*kg*K/m*K*kg*J

$\boldsymbol{\alpha_{high} = 9.23*10^{-4}\ m^2/s}$

4. Calculate conduction thickness needed to keep back-face safe
    - Sufficiently low T of back wall
5. Calculate thermal penetration depth
    - For given t_{burn}
    - Transient consideration

$\delta = (\alpha*t_{burn})^{1/2}$

6. Estimate ablation/recession
    - $\Delta_{ablate} = r(dot)_{erosion} * t_{burn}$

$r(dot)_{erosion}$ (for LOX/Kerosine O/F) = $6*10^{-5}$ m/s

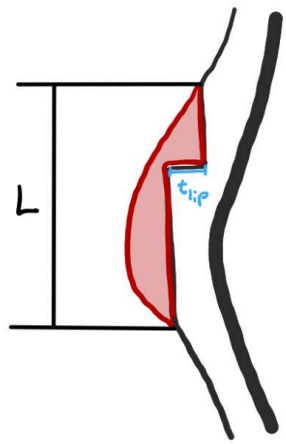
Δ_{ablate} = 0.0006m = 0.6mm

This means that a graphite insert would ablate by 0.6mm for a 10 second burn time.

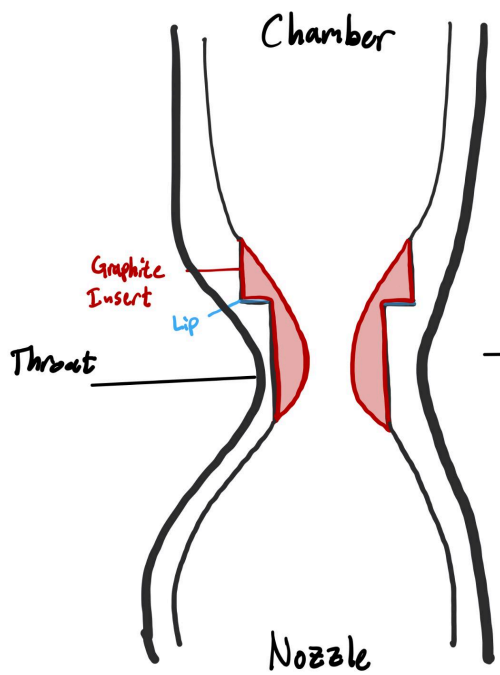
7. Add mechanical/support thickness
    - Lip thickness & safety margin
    - *That sum is the initial graphite thickness/length at throat*

Length of graphite insert must be enough to rest on mechanical lip in nozzle throat design

$$L = L_{cond} + \Delta_{abl} + t_{lip} + L_{safety}$$

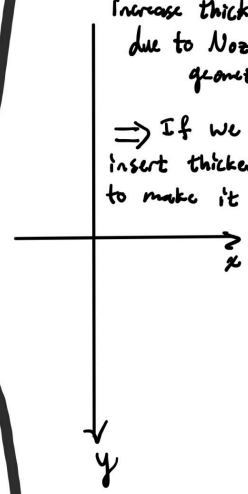


- $L_{cond} = \frac{k(T_h - T_b)}{q''}$
  - $k \equiv$ Thermal conductivity of Graphite
  - $T_h \equiv$ Temp of graphite on side facing GAS
  - $T_b \equiv$ Temp of graphite on side facing CHAMBER WALL
  - $q'' \equiv$ Surface heat flux at throat
- $\Delta_{abl} = \dot{r}_{erosion} \cdot t_{burn}$
  - $\dot{r}_{erosion} \equiv$ Graphite erosion rate
  - $t_{burn} \equiv$ Burn time
- $t_{lip} \equiv$ thickness of lip
- $L_{safety} \equiv$ Safe extra length for unforseen erosion ($\sim$ 2-7mm)



An increase in $L\hat{y}$ will inherently increase thickness $\hat{x}$ due to Nozzle geometry.

$\Rightarrow$ If we must make the insert thicker, it is reasonable to make it longer.



- Throat *axial* length t (the minimum-diameter barrel length) — often ~0.3–1.0×d_t. Shorter helps reduce pressure loss but may erode faster or be less stable.
- Minimum graphite thickness at throat (axial, from hot face to the back seat): **typical** 5–40 mm depending on engine size, flux, and burn time. Small, short-burn engines at low flux: ~5–15 mm. High flux / long burns require much thicker inserts.
- Back-face temperature goal: keep metal carrier below ~400–800°C depending on alloy; pick conservative 300–500°C if using stainless or copper alloys.
- Radial clearance between graphite bore and metal carrier bore: **~1–3 mm** to avoid metal intrusion.
- Lip thickness (minimum hot-face edge): **2–5 mm** to prevent chipping (depends on machining and loading).
- Seat geometry: use tapered seats (3–8°) that allow thermal slip and prevent point stresses.

Which Graphite do we use?

Fine-Grain Isotropic Graphite

Pros:

- Isotropic thermal expansion
- High k value
- Easily machine-able
- Accessible

Cons:

- Brittle
- Becomes less stable ~2500°C

Where to find:

- Poco Graphite (AXF-5Q), (AXM-5Q)
- Toyotanso (IG-11)

Resin-Impregnated Graphite

Pros:

- Stable near ~3000°C
- Highly dense

Cons:

- Expensive
- Hard to machine
- Lower k value than Fine-Grain Isotropic
- Fewer suppliers

Where to find:

- Poco Graphite (DFP-1)